ISL Reader: A Two-Megabyte Vision Transformer, a Super-Resolution GAN Front End and an Integrated Interface for Indian Sign Language, Evaluated on Unseen Recording Sessions

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Abstract

Word-level Indian Sign Language (ISL) recognisers in the literature report accuracies above 94 %, usually on random splits that place clips from one recording session on both sides of the train–test boundary. We describe ISL Reader, a working recognition application built around ISL-ViT-Tiny, a factorised video Vision Transformer that reads cropped left-hand, right-hand and face streams together with hand landmarks, and we evaluate it only on recording sessions withheld from training. On the 472 held-out clips of the 50 deployed words, the shipped 4-bit model (3.89 M parameters, 1.97 MiB) reaches 85.9 % top-1 and 96.4 % top-5 accuracy, the mean of three seeds; for comparison, a pixel-only model that scored 75.6 % on unseen sessions scored 98.5 % on a random split. Because a wearable camera may deliver low-resolution video, we placed a super-resolution generative adversarial network (GAN) in front of the hand detector and compared it with bicubic upscaling and with the same generator trained without the adversarial loss. On 180p video, learned super-resolution raised top-1 accuracy from 61.0 % to between 75.4 % and 76.0 %, but no adversarial variant, including one trained against a differentiable hand-landmark proxy, was reliably better than the non-adversarial control. A seven-screen web interface connects the models to a camera, uploaded files or built-in demonstration clips, answers only above a confidence threshold, and logs corrections. Automated browser tests found no runtime errors, and the interface's answers matched the offline evaluation on every clip checked. Pixel-only checkpoints scored at chance on a second corpus, so these results describe studio video of the INCLUDE kind.

Keywords: Indian Sign Language · Vision Transformer · super-resolution · generative adversarial network · signer-independent evaluation · quantisation-aware training · user interface

1 Introduction

Indian Sign Language is the first language of a large deaf community, and few hearing people can read it. A wearable assistant that watches a signer and shows the word to a hearing listener would remove part of that barrier. This project builds the recognition side of such a device and, for this review, the application around it: given a short clip of one sign, the system names the word, says how sure it is, and lets the user confirm, correct or reject the answer. The target hardware is a pair of smart glasses, so the classifier has to fit in about 2 MB.

Three properties make the task hard. ISL is largely two-handed, and a word is defined by handshape, motion and location together, so the same handshape at the forehead and at the chest can mean different things. Labelled data is scarce: INCLUDE [16], the main public word-level corpus, holds 4,257 usable clips over 262 words, about 15 per word, whereas vision transformers are normally trained on orders of magnitude more. And a device worn on the head sees the signer through a small, often compressed camera, so the input is likely to be lower in resolution than the 1080p studio video the model learned from.

A fourth difficulty is in how results are usually measured. INCLUDE was recorded as consecutive takes of each word by the same person in the same room. A random split therefore puts near-duplicate takes on both sides of the boundary, and a model can score highly by recognising the session rather than the sign. A deployed device only ever meets people and rooms it has not seen, so we test on whole recording sessions held out from training.

Earlier reviews of this project produced the classifier and its evaluation protocol. Review 3 asked for the models to be integrated with a user interface and the results compared. This paper therefore covers the full system and makes four contributions:

1. **A complete, tested application.** A web interface with seven screens runs any trained classifier checkpoint and either upscaling front end, and its answers reproduce the offline evaluation clip for clip.
2. **A compact classifier measured honestly.** A 3.89 M-parameter factorised transformer, compressed by quantisation-aware training to a 1.97 MiB file, reaches 85.9 % top-1 on unseen sessions; we compare it with pixel-only, full-precision and post-training-quantised variants.

3. **A controlled study of a GAN front end.** We compare bicubic upscaling, a non-adversarial learned upscaler, three adversarially trained generators and a task-aware generator on degraded test video, with paired tests and a rule fixed before the runs.
4. **A diagnosis of what limits it.** By swapping clean and degraded inputs, we show that after super-resolution most of the remaining loss comes from the hand landmarks, not the pixels, and that tuning the detector does not recover it.

2 Related work

2.1 Transformers for sign recognition

Vision transformers [19] have moved quickly into sign recognition. Ashfaq et al. [3] add an enhanced feed-forward head to a ViT and report 99.44–99.90 % on four static hand-gesture datasets. Shawon et al. [8] compare VideoMAE, ViViT and TimeSformer on word-level corpora and reach 96.9 % on BdSLW60 with held-out users, using models far too large for a wearable. Abdi and Barzegar Touchahi [5] report that a small ViT lost 2.1 points in murky water against 12.1 points for ResNet-18, an indication that attention degrades more gracefully than convolution when the input is corrupted, which is the situation our low-resolution front end addresses. Rubaiyeat et al. [4] apply a transformer to anatomically anchored landmarks. Our classifier follows the factorised-encoder design of ViViT [21] and starts its spatial stage from DeiT-Tiny [20].

2.2 Compact recognisers for edge devices

TinyMSLR [1] fuses ConvNeXt-Tiny and Swin features and distils from two teachers, reaching 99.01 % on 20 classes with under 2.7 M parameters and 24 ms per sample on a CPU. Venkatappa et al. [2] quantise a landmark transformer for 33 Kannada signs to 1,097.77 KB and run it in 16.2 ms on a phone at 94.19 %. That model is close to our size budget, but it discards appearance by using landmarks alone and quantises after training; we keep the image crops and quantise during training.

2.3 Indian Sign Language and evaluation practice

Nedungadi et al. [6] built a CNN–ViT conversational agent for ISL e-governance at 97.5 %, Kaliyaperumal et al. [7] recognise two-handed dynamic ISL words with a convolutional transformer, and Damdoo et al. released the sentence-level ISH-NEWS corpus [9] and surveyed the field, identifying data scarcity and evaluation practice as open problems [10]. None of these reports signer- or session-disjoint results. Newer corpora ship a signer split: Isharah [11] defines signer-independent benchmarks, and TSL-ONE-S [12] reports signer-independent top-1 of 79.69 % (I3D) and 80.83 % (SPOTER) on 184 Thai glosses, the nearest like-for-like comparison to our setting. INCLUDE has no signer labels, so we reconstruct sessions ourselves (Section 3.3).

2.4 Super-resolution GANs

Generative adversarial networks [27] train a generator against a discriminator that learns to tell its output from real data. SRGAN [28] applied this to single-image super-resolution, pairing an adversarial loss with a perceptual loss computed on VGG features [33, 34]; ESRGAN [29] refined the architecture and introduced the relativistic discriminator [32] to the task. Real-ESRGAN [30] targets real rather than synthetic low-resolution images by training on a randomised second-order degradation pipeline, uses a U-Net discriminator [31] with spectral normalisation [35], and provides a compact VGG-style generator that we adopt. These methods optimise for images that look realistic to people; whether they help a downstream recogniser is a separate question, and the answer depends on what the recogniser relies on. We return to this in Section 4.4, where we add a loss computed by a differentiable stand-in for the hand-landmark detector, built in the style of heatmap pose estimators [36, 37].

3 Methodology

3.1 System overview

Fig. 1 shows how a sign becomes a word, using real frames from a held-out test clip. Fig. 2 shows the deployed system. The browser records or loads a clip and posts it to a Python server. The server decodes 32 frames, upscales them if the video is below 720p, locates the hands, face and body with MediaPipe Holistic [23], cuts out three crops per frame and extracts 21 joints per hand, and passes 16 frames of this to the classifier. The classifier scores six views of the clip; the server returns the five most likely words with their probabilities and how each crop was obtained. The browser shows the answer only if the top probability exceeds a threshold, and otherwise asks the user to pick from the shortlist.

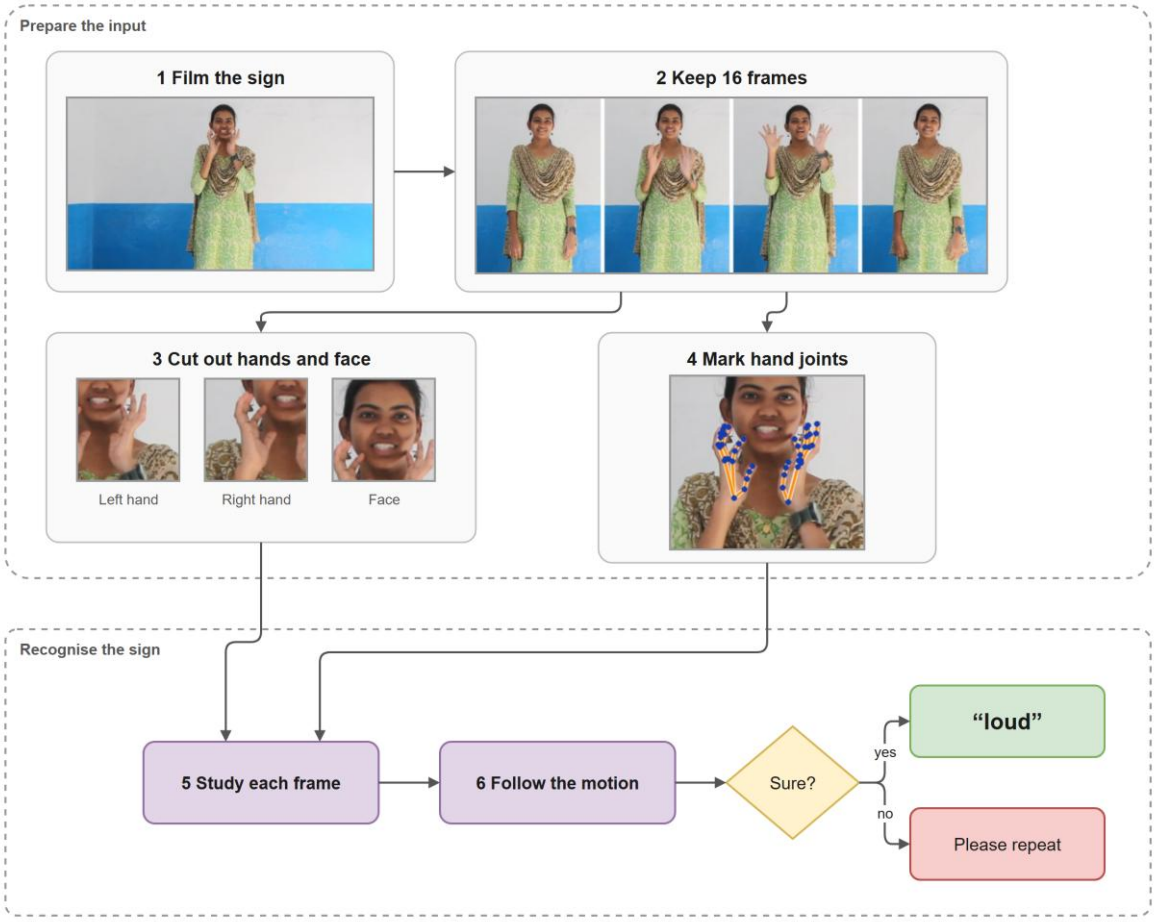


Fig. 1 How ISL Reader reads a sign. Frames, crops and detected hand joints are real data from a held-out test clip of the word "loud".

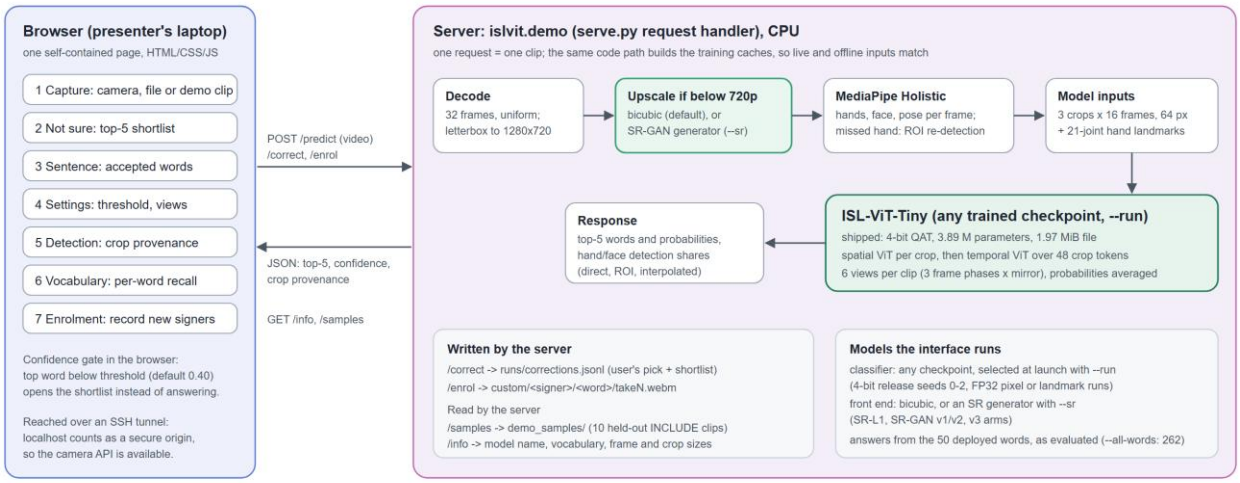


Fig. 2 Architecture of the integrated application. Any trained classifier checkpoint and either upscaling front end can be selected when the server starts.

3.2 Dataset

Table 1 summarises the data. INCLUDE [16] provides the labelled clips, recorded at 1920 × 1080 and 25 frames per second in one studio and released under CC-BY-4.0. We train on all 262 words and restrict the output to the 50 deployed words at test time; a smaller training vocabulary would only remove data, because the number of clips per word is fixed. iSign [18] supplies 18,000 unlabelled clips for self-supervised pretraining, and CISLR [17], recorded with different signers and cameras at 300 × 300 pixels, serves as a cross-corpus test.

Table 1 Data used in this work.

Item	Value
INCLUDE clips used	4,257 clips, 262 words, 15 categories; 1920 × 1080, 25 fps
Recording sessions	13, reconstructed from camera clip numbers
Training split	2,845 clips from training sessions
Test split	1,010 clips from held-out sessions; 472 of them belong to the 50 deployed words
Excluded	402 held-out-session clips of words with fewer than six training clips
Validation sessions (GAN development only)	888 clips, a subset of the training sessions held out from all generator training
Unlabelled pretraining	18,000 iSign clips
Cross-corpus test	CISLR, 609 clips, 300 × 300 pixels

3.3 Evaluation protocol

Every INCLUDE file is named after the camera's running counter. Consecutive numbers are consecutive takes; large jumps separate sessions. Clustering the counter over all 4,257 clips gives 13 session blocks, and whole blocks are held out for testing (Fig. 3). The split builder asserts that no counter value appears on both sides. The same 472-clip, 50-word test set is used for every result in this paper. Because sessions are inferred, the protocol approximates signer independence rather than guaranteeing it.

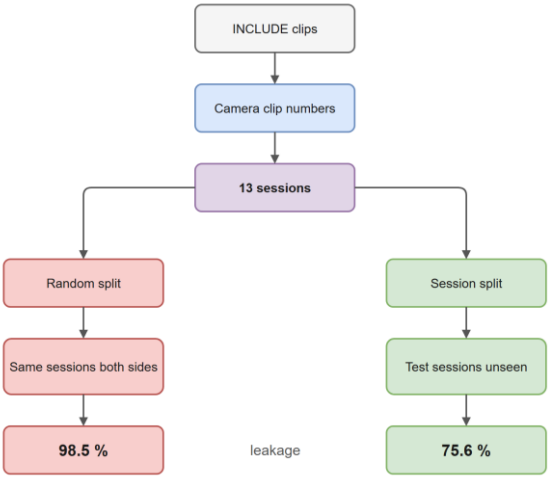


Fig. 3 Building the evaluation split from camera clip numbers. The same pixel-only model scores 98.5 % on a random split and 75.6 % on unseen sessions.

We report top-1 and top-5 accuracy, with balanced accuracy where per-word counts differ. Each result is the mean of three independently trained seeds; we never report the best seed, since picking it would amount to selecting on the test set. Two front ends or models are compared on identical clips with McNemar's exact test [26], pooling discordant clips over the three seeds. Before the super-resolution runs we fixed one rule: a change counts as an improvement only if the mean rises, all three seeds move in the same direction and the pooled p-value is below 0.05; the same rule in reverse marks a change as harmful.

3.4 Input pipeline and hand landmarks

Each clip is sampled uniformly to 32 frames and letterboxed to 1280 × 720. MediaPipe Holistic finds the hands, face and body pose in each frame. When a hand is missed, a second pass runs MediaPipe's hand model on a forearm-sized region around the pose estimate, upscaled to 256 pixels; if that also fails, the nearest real box is reused and marked as interpolated. Crops are stored at 128 pixels and resized to 64 for the model. MediaPipe's detectors keep state between videos, which changed hand presence on 3–6 % of frames when the same clip was processed twice; the detectors are therefore rebuilt for every video, after which repeated runs agree exactly.

The detector's 21 joints per hand are also passed to the model (Fig. 4). Handshape is the joints relative to the wrist divided by the hand's extent, so it does not depend on where the hand is or how far away. Location is the wrist relative to the shoulder midpoint in shoulder widths, and body pose uses the nose, shoulders, elbows and wrists in the same frame. A

two-layer network adds these features to the corresponding crop token; its last layer starts at zero, so the model begins training as the pixel-only model it extends.

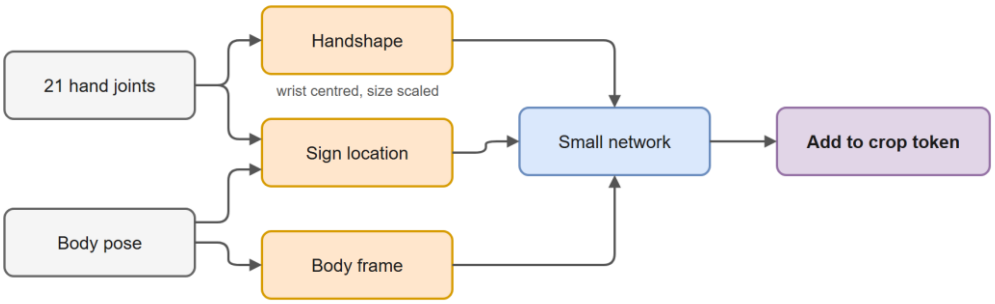


Fig. 4 Hand-landmark features. Joints are centred on the wrist and scaled by hand size before a small network adds them to the crop token.

3.5 Classifier architecture

ISL-ViT-Tiny (Fig. 5, Table 2) is a factorised video transformer. The spatial stage, a four-block ViT, encodes one 64 × 64 crop at a time as 16 patches plus a class token, with the same weights for all 48 crops of a clip (16 frames × 3 streams). Each crop's class token then receives a stream embedding, a frame embedding, the crop box's position and size, a learned flag for interpolated boxes and the landmark features. Cropping removes where the hand is, and in ISL location separates words, so the box geometry puts that information back. A second four-block transformer attends over the 48 crop tokens and a class token, and a linear layer produces the word scores. Attending jointly over every patch would mean 816 tokens in one sequence; the factorised model attends over 17 tokens per crop and then 49 across the clip.

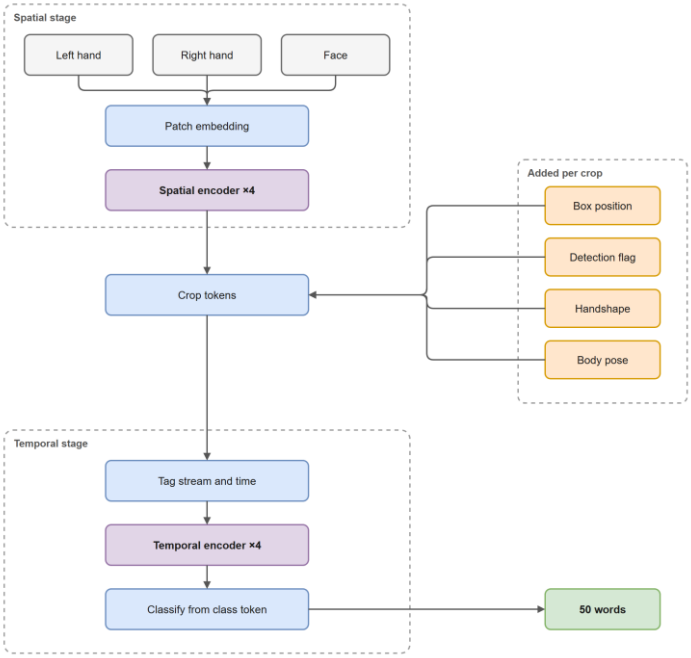


Fig. 5 ISL-ViT-Tiny. The spatial stage runs once per crop with shared weights; the orange inputs are added to every crop token before the temporal stage.

Table 2 Architecture of the deployed classifier.

Component	Setting
Input per clip	16 frames × 3 crops × 64 × 64 RGB, box geometry, detection flags, hand landmarks
Patches per crop	16 × 16 pixels, 16 patches + class token
Spatial stage	4 blocks, width 192, 3 heads, MLP 768, shared across 48 crops
Temporal stage	4 blocks, width 192, 3 heads, over 48 crop tokens + class token
Output	Linear 192 → 262, restricted to the 50 deployed words at test time
Parameters	3,894,214 (91 k of them for the landmark features)

3.6 Training and compression

Training has two phases (Fig. 6). Both transformer stages are first pretrained on the 18,000 unlabelled iSign clips by masked image modelling in the style of SimMIM [22], reconstructing 75 % of hidden patches. Hands in iSign are small in the frame and become blurry when cropped, so fine-tuning degrades each INCLUDE crop to a random lower resolution and back (resolution jitter 0.5) to match. The full model is then trained on the 2,845 training clips for a fixed 1,000 epochs with AdamW (learning rate 5×10^{-4} with 10 warm-up epochs and cosine decay, encoder rate scaled by 0.1, batch 64, weight decay 0.05), label smoothing 0.1, mixup ($\alpha = 0.2$, probability 0.5), stochastic depth 0.1, stream dropout 0.15, colour jitter 0.4, greyscale 0.15, horizontal flips that swap the hand streams, gradient clipping at 1.0 and an exponential moving average of the weights. The final epoch is kept; no checkpoint is chosen by test accuracy.

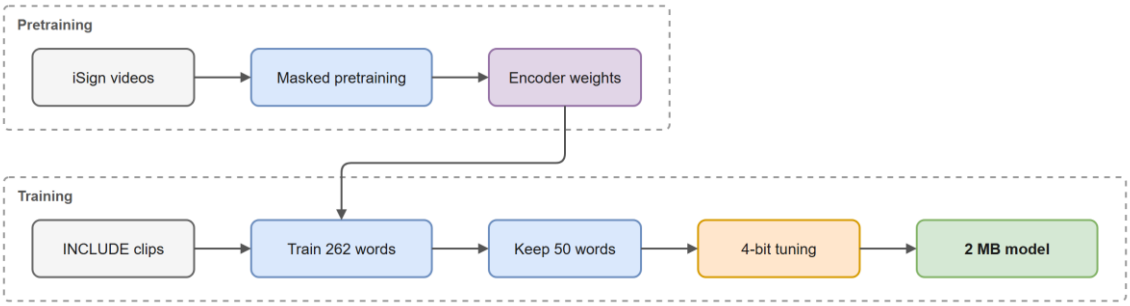


Fig. 6 Training pipeline, from unlabelled pretraining to the 2 MB deployed model.

At 32-bit precision the model occupies 14.9 MB. We store the weight matrices at 4 bits in groups of 128 with a 16-bit scale per group, keeping embeddings and small tensors at 16 bits. Rounding after training lost 3.1 points on average, so we fine-tune for a fixed 300 epochs with the rounding inside the forward pass and a straight-through estimator [25] in the backward pass, following Jacob et al. [24] (Fig. 7). The training loop and the exporter share one tensor-selection function and one rounding function, and reloading the saved file reproduces every weight bit for bit. The shipped file is 2,069,700 bytes.



Fig. 7 Quantisation-aware training. The dashed loop runs for 300 epochs.

3.7 Super-resolution GAN front end

Every clip reaches the hand detector through one function, which resizes video below 720p to the 720p working height. That resize was bicubic. The GAN replaces it, and only it, when the server is started with a generator file; video at 720p or above never passes through the generator, so results on the original 1080p data cannot change. Fig. 8 shows the networks and losses.

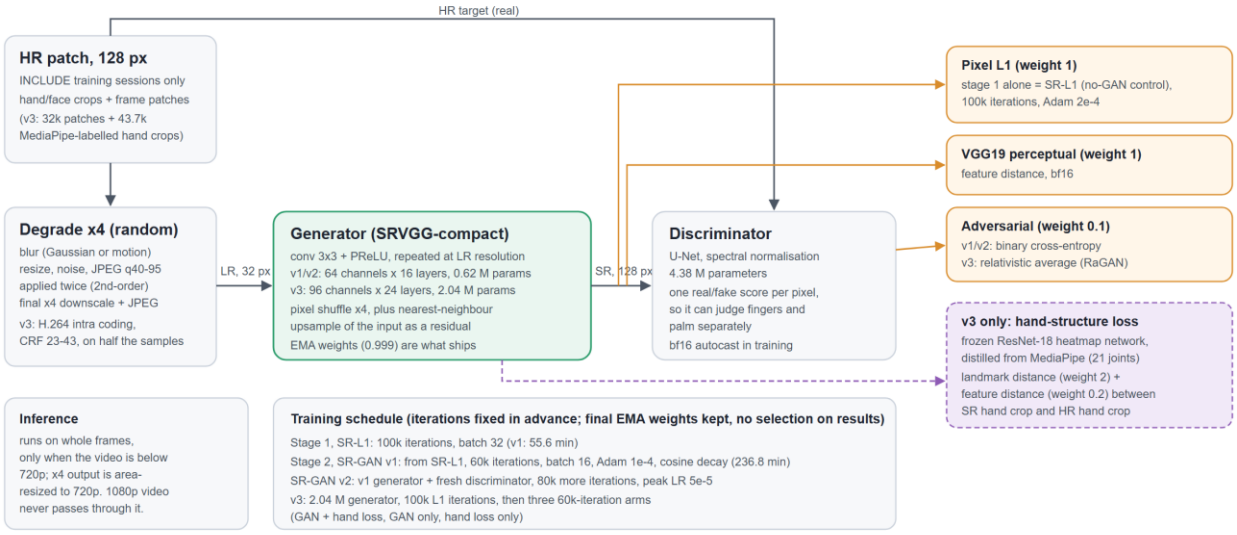


Fig. 8 The super-resolution GAN: training data and degradation, generator, discriminator, losses and schedule.

Generator and discriminator. The generator is Real-ESRGAN's compact VGG-style network [30]: 3 × 3 convolutions with PReLU activations at the low resolution, a ×4 pixel shuffle, and a nearest-neighbour upsample of the input added as a residual, so an untrained network falls back to plain upscaling. Versions 1 and 2 use 64 channels and 16 layers (621,424 parameters); version 3 uses 96 channels and 24 layers (2.04 M). The discriminator is a U-Net with spectral normalisation that scores realism per pixel (4.38 M parameters), so it can penalise an implausible fingertip without penalising the palm next to it.

Training data and losses. High-resolution patches of 128 pixels come only from INCLUDE training sessions, half from detected hand and face crops and half from whole frames; a check refuses any clip from the test sessions. Each patch is degraded at random by blur, resizing, noise and JPEG compression applied twice, then downscaled by four. Stage 1 trains the generator with a pixel L1 loss for a fixed 100,000 iterations; this network, called SR-L1, is also the control that separates what learned upscaling gives from what the adversarial loss adds. Stage 2 continues from SR-L1 with the L1 loss, a VGG19 perceptual loss and an adversarial loss weighted 0.1, for a fixed number of iterations. SR-GAN version 1 ran 60,000 iterations; version 2 continued from version 1 with a fresh discriminator for 80,000 more at half the learning rate. Both keep the final exponential-moving-average weights.

Study design. The 472 test clips were degraded inside the frame reader, before detection, to three levels: 360p with JPEG quality 60, 240p with quality 50 and 180p with quality 40. Each level was passed through each front end, and the three released 4-bit models and their 32-bit counterparts were scored without retraining. A second arm fine-tuned the 32-bit models for 150 epochs on 360p training video passed through either bicubic or SR-GAN version 2, to ask whether the GAN still helps once the classifier has adapted. We also measured hand detection against the clean video, including hands detected in degraded video that the clean video does not have, which is how a GAN that invents hands would show up.

A task-aware generator. Section 4.4 shows that, after super-resolution, most of the remaining error comes from the hand landmarks. MediaPipe is not differentiable, so we trained a stand-in for it: a ResNet-18 [36] whose features are expanded by three transposed convolutions into 21 heatmaps, read out by soft-argmax [37]. It learned from 43,682 hand crops whose MediaPipe landmarks we could verify belonged to the crop, from training sessions only. On 2,466 crops from the held-out validation sessions its mean landmark error is 0.0215 of the crop width, with 91.4 % of points within 0.05; for reference, MediaPipe re-run on the same crops differs from its own stored labels by 0.041 on average. Version 3 of the generator was trained with this network frozen, adding the distance between its landmark reading of the generated crop and of the original (weight 2) and between their features (weight 0.2), together with a relativistic adversarial loss and H.264 intra-frame compression in the degradation. Three arms were trained from one 100,000-iteration L1 network: adversarial with the hand losses, adversarial without them, and the hand losses without the adversarial term. Development decisions used 200 clips from the 888 validation-session clips, never the test set.

3.8 User interface and integration

The interface is a single self-contained web page served by the recognition server (Fig. 2). It has seven screens. Capture records three seconds from the camera, accepts a dropped or chosen video, or plays one of ten built-in clips, and shows the answer inside a ring whose clarity reflects the model's confidence. Not sure lists the five candidates when the top word

falls below the threshold, and a pick is written to a corrections log as a labelled example. Sentence collects accepted words. Settings sets the threshold and the number of views. Detection shows, per clip, how many hand and face crops came from real detections, from the second-pass search or from interpolation. Vocabulary shows per-word recall on the test sessions. Enrolment records three takes per word from new signers into a folder the training pipeline reads.

Three integration details matter for the comparisons in this paper. The server runs any trained checkpoint, chosen at start-up, and either upscaling front end. Its answers are restricted to the 50 deployed words in exactly the way the evaluation restricts them: the six views' probabilities are averaged over all 262 words, the other 212 are set to zero, and the top remaining word is the answer. Without this step the live application would answer from all 262 words, which for the same models scores 72.0 % rather than 85.9 %. Finally, the presenter reaches the server through an SSH tunnel, which also makes the page a secure origin so the browser allows the camera.

The ten demonstration clips are held-out test clips transcoded to 720p H.264, the same files the model reads. Nine are words that all three released models classify correctly, each confirmed against the running server (confidence 66–97 %). The tenth is a look-alike word, "wide", chosen because the model is unsure of it (top probability 7.3 %) while the right word is still in its shortlist, so the demonstration includes the Not-sure path. The clips illustrate the interface; they are not a measure of accuracy.

3.9 Testing the interface with the models

We tested the interface at three levels. Automated browser tests (Playwright driving Chromium) opened all seven screens at 1366 and 1920 pixels wide and recorded console errors, layout measurements and the timing of the screen transitions. End-to-end tests posted held-out clips to the live server through the interface and checked the answer, the confidence and the detection breakdown shown on screen. A consistency test called the server's classification code on held-out test clips and compared each answer with the offline evaluation's result for the same clip and model.

4 Results and discussion

4.1 Leakage in the standard split

A pixel-only model scored 98.5 % on a random split of the 50 words and 75.6 % on unseen sessions (Fig. 9). These two figures come from the project's earlier controlled runs, with the model and recipe held fixed. The 22.9-point difference is what the model gains from having seen the same person, room and take before, and it is the reason every other result here uses held-out sessions.

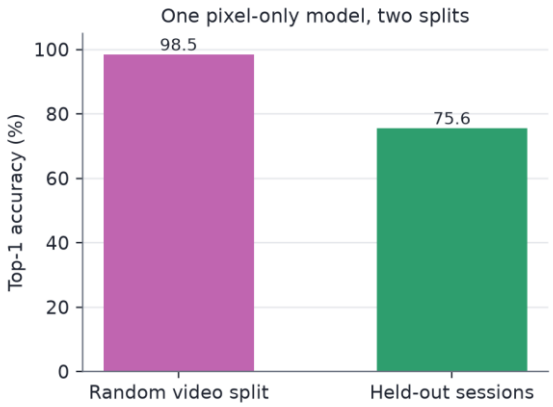


Fig. 9 One pixel-only model, two splits.

4.2 Classifier variants

Table 3 and Fig. 10 compare the model variants on the same 472 clips. Adding the hand landmarks to the pixel-only model raised top-1 from 73.7 % to 86.5 %; on identical clips the landmark model in the earlier two-seed comparison fixed far more clips than it broke (McNemar p below 10^{-9} for each seed). Rounding the landmark model to 4 bits after training cost 3.1 points, and quantisation-aware training recovered all but 0.6 of them, at about a seventh of the file size. The shipped model is 10.9 points above the project's accuracy target of 75 % at its size target of about 2 MB.

Table 3 Classifier variants on the 472 held-out clips of the 50 deployed words, six-view averaging. Means of three seeds, individual seeds in brackets.

Model	File size	Top-1 (%)	Top-5 (%)
Pixel-only, 32-bit	14.55 MB	73.7 (75.0 / 73.7 / 72.2)	93.5
+ landmarks, 32-bit	14.9 MB	86.5 (87.3 / 86.9 / 85.4)	96.5
+ landmarks, 4-bit after training	≈ 2 MB	83.4 (83.1 / 85.2 / 82.0)	—
+ landmarks, 4-bit quantisation-aware (shipped)	**1.97 MiB**	**85.9 (84.5 / 87.5 / 85.8)**	**96.4**
Shipped model, answering from all 262 words	1.97 MiB	72.0 (69.7 / 75.8 / 70.3)	92.7

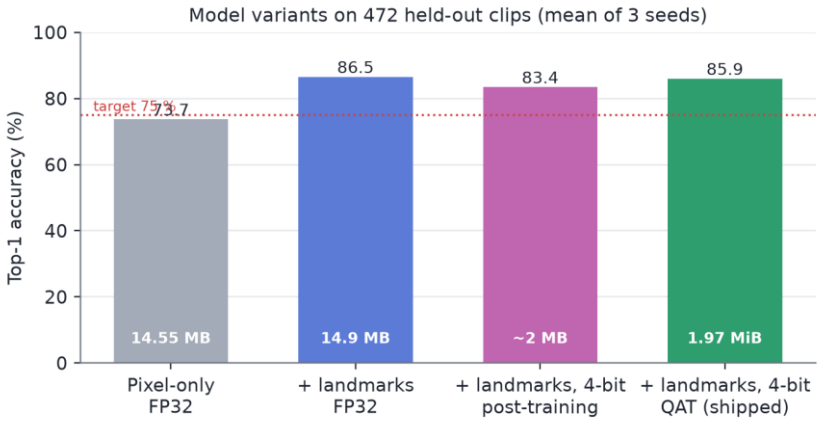


Fig. 10 Top-1 accuracy of the classifier variants. The dotted line marks the 75 % target.

Two figures put these numbers in context. The nearest like-for-like result we found, signer-independent recognition of 184 Thai signs [12], is 79.69–80.83 %, which suggests that honest accuracy on unseen people is in the high 70s to mid 80s rather than above 95 %. And the shipped model's top-5 accuracy of 96.4 % matters for the interface, because when the model is unsure the right word is usually among the five candidates the Not-sure screen offers. For an earlier pixel-only ensemble, a threshold of 0.4 answered 75.2 % of clips with 88.7 % of those answers correct; we have not re-measured that curve for the landmark model, so the interface's Settings screen labels it as an older measurement.

4.3 The GAN front end

Fig. 11 and Table 4 show the main comparison. Learned super-resolution helped at every degraded level under the pre-registered rule. With the frozen 4-bit models, SR-GAN version 2 raised accuracy over bicubic by 1.9 points at 360p, 6.4 at 240p and 14.5 at 180p, closing 40–64 % of the gap to clean video, and the same front end helped the fine-tuned models by 5.2 points at 240p and 12.5 at 180p (the 1.3-point gain at 360p was not significant). At 1080p the front end is never used, and the clean result stayed at 85.9 %.

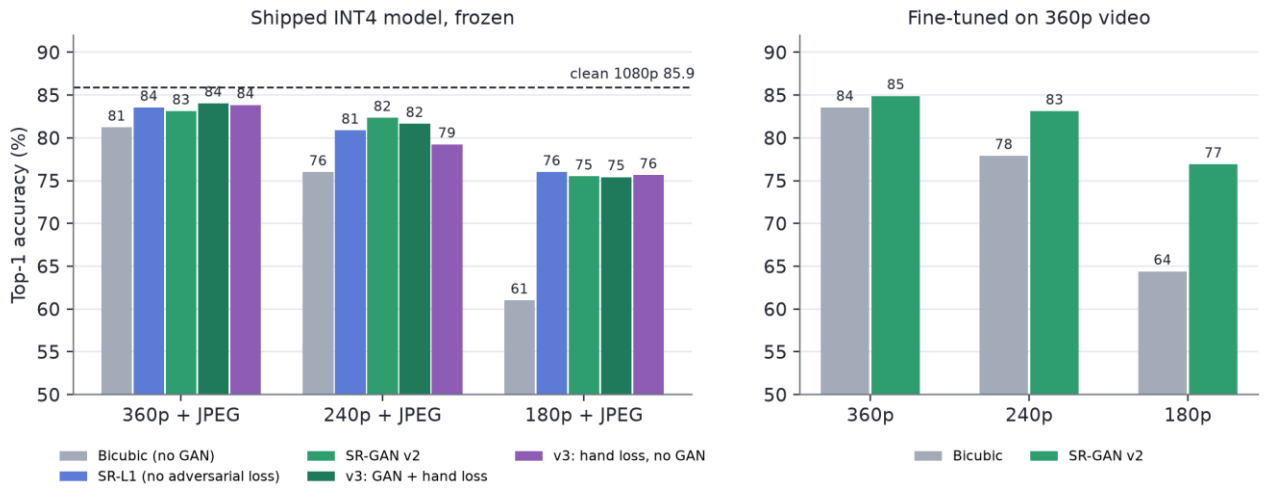


Fig. 11 Top-1 accuracy on degraded test video. Left: the shipped 4-bit models, unchanged. Right: 32-bit models fine-tuned on 360p video through each front end.

Table 4 Paired comparisons on the 472 test clips, frozen 4-bit models. Δ is the mean top-1 change over three seeds; clips gained and lost are pooled over seeds; the verdict applies the pre-registered rule.

Input	Comparison	Δ (points)	gained / lost	p	Verdict
360p	SR-GAN v2 vs bicubic	+1.9	73 / 46	0.017	helps
360p	SR-GAN v2 vs SR-L1	−0.4	28 / 34	0.53	none
240p	SR-GAN v2 vs bicubic	+6.4	142 / 52	8×10^{-11}	helps
240p	SR-GAN v2 vs SR-L1	+1.5	49 / 28	0.022	helps
180p	SR-GAN v2 vs bicubic	+14.5	265 / 60	7×10^{-32}	helps
180p	SR-GAN v2 vs SR-L1	−0.5	56 / 63	0.58	none
360p	SR-GAN v2 vs v1	−1.2	16 / 33	0.021	hurts
240p	v3 GAN + hand loss vs SR-GAN v2	−0.7	38 / 48	0.33	none
180p	v3 GAN + hand loss vs SR-GAN v2	−0.1	62 / 63	1.0	none
240p	v3 hand loss, no GAN vs SR-GAN v2	−3.1	24 / 68	5×10^{-6}	hurts
240p	v3 GAN + hand loss vs v3 hand loss, no GAN	+2.4	68 / 34	0.001	helps

The adversarial loss itself added little. SR-GAN version 2 beat the non-adversarial SR-L1 only at 240p (+1.5 points); at 360p and 180p the difference was within noise. Training the GAN longer did not help either: version 2, with 140,000 adversarial iterations in all, was not better than version 1 at any level and was 1.2 points worse at 360p. The adversarial loss did change the images, as intended: on held-out patches at 180p it doubled crop sharpness over SR-L1 (variance of the Laplacian 32 against 16, originals 73) while lowering PSNR from 31.83 to 31.09 dB, the familiar perception–distortion trade-off. Sharper texture did not translate into higher accuracy.

Detection explains where the accuracy came from. At 180p the bicubic front end found the left and right hand in 51.0 % and 53.8 % of frames and the face in 33.5 %; with SR-GAN version 2 the figures were 68.6 %, 71.9 % and 99.9 %, and the share of hands present in clean video but missed fell from 39.3 % to 22.1 %. Hands detected in degraded video but absent from the clean video rose only from 1.9 % to 2.6 %, so the GAN did not obtain its gain by inventing hands. On CISLR, which is natively 300 pixels, the front end left left-hand detection unchanged (25.0 % with bicubic, 24.3 % with SR-GAN); Section 4.5 shows why.

4.4 Where the remaining loss comes from

To locate the loss that super-resolution leaves, we kept one input clean and degraded the other (Fig. 12). At 180p, giving the model clean pixels but degraded landmarks scored 78.2 % with SR-GAN version 2; giving it degraded pixels but clean landmarks scored 83.1 %. Of the 10.4 points still missing, about 7.7 are therefore on the landmark side and 2.8 on the pixel side, and the two add up. With bicubic upscaling both sides lost heavily (12.2 and 6.9 points) and the losses compounded. Super-resolution has repaired most of the pixel damage; what remains is mostly in where the detector places the 21 joints.

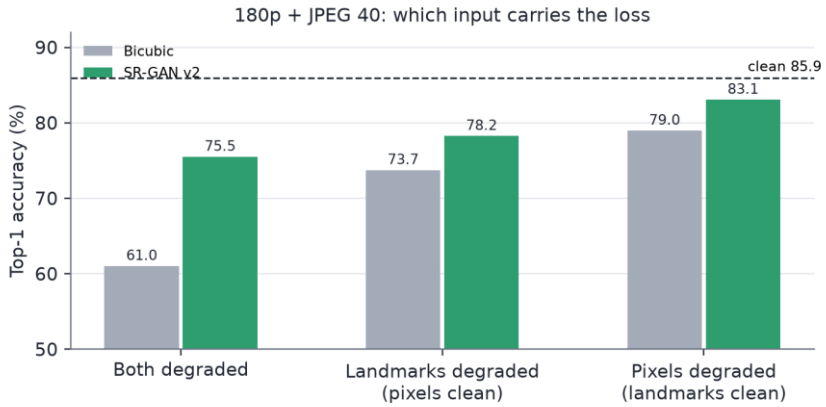


Fig. 12 180p, frozen 4-bit models: accuracy when only the landmarks, or only the pixels, come from degraded video.

Tuning the detector does not recover this. Re-running landmark extraction on the same super-resolved frames with lower confidence thresholds, a larger or multi-scale search region and gap filling raised the share of hand frames found from 70.3 % to as high as 99.2 %, above the clean video's 89.7 %, but accuracy moved by at most 0.8 points, because the misplaced hands rose from 4.7 % to between 7.1 % and 14.1 %. The limit is the quality of the landmarks, not whether a hand is found.

This is the motivation for version 3. On the validation sessions it did what it was designed to do at the detection level (Fig. 13): the adversarial arm with hand losses found hands in 73.3 % of 180p frames against 63.3 % for version 2, and the hand-loss arm without the adversarial term had the lowest handshape error (0.99 against 1.30, in hundredths of the frame height). On the test set, however, neither arm was reliably better than version 2 or SR-L1 at any level (Table 4), and the hand-loss arm without the adversarial term was 3.1 points worse at 240p. Better detection statistics on validation did not become better classification. One possible reason, which we have not tested, is that the frozen classifier was trained on MediaPipe's landmarks from clean video and responds to the detector's errors in ways that our proxy distance does not capture; fine-tuning the classifier on the version 3 front end would test this.

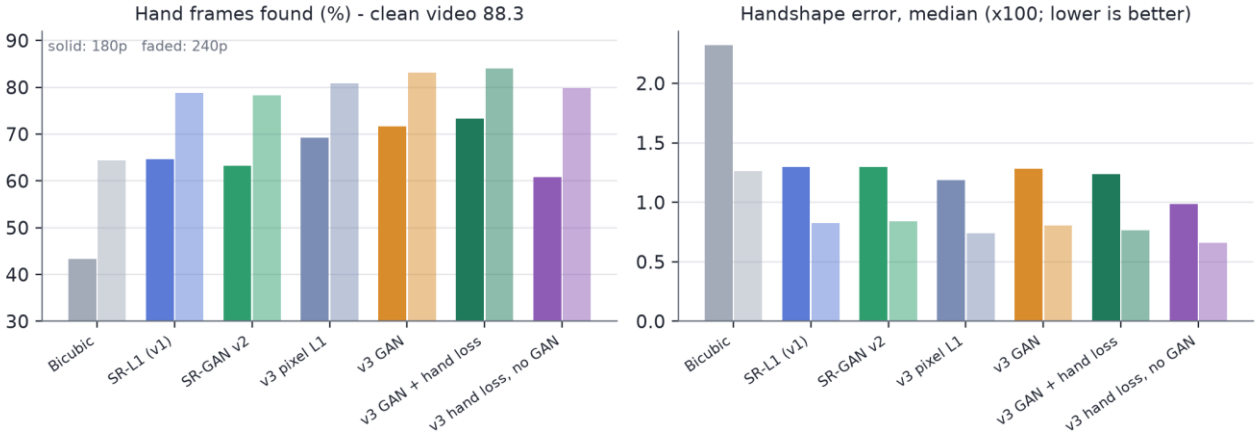


Fig. 13 Validation sessions, 200 clips, MediaPipe on the super-resolved video against the clean video. Solid bars 180p, faded 240p.

4.5 The second corpus

Pixel-only checkpoints that reached up to 42 % on INCLUDE's unseen sessions in earlier experiments scored 0.3–1.5 % on CISLR, which is chance. Super-resolution does not change this, because CISLR's problem is not resolution. Its signers are framed tightly, with hands about 14 % of the frame width against 8.5 % in INCLUDE, yet MediaPipe's pose model, which expects a full upper body, reports a visible wrist in only 22 % (left) and 42 % (right) of frames. When the wrist is found, the hand is found 79–85 % of the time. A hand detector run directly on the whole frame found a hand in 47 % of 318 sampled frames at confidence 0.3, against 40 % for the pose-led path. The fix for CISLR lies in detection, not in upscaling.

4.6 Interface tests

Table 5 lists the interface tests. All seven screens rendered at both widths with no browser console errors. Sent to the live server, the demonstration clips gave the expected outcomes: confident correct answers for the nine reliable clips and, for

the look-alike clip, the Not-sure shortlist with the right word first; four of them were also played through the Capture screen, with the same results (Figs. 14 and 15). The consistency test agreed with the offline evaluation on all 8 held-out clips checked, including the 2 the model gets wrong. On a desktop CPU a clip took between 5 and 9 seconds to read with six views. With the GAN front end enabled, a 180p copy of a demonstration clip was read correctly.

Table 5 Tests of the interface with the models.

Test	Method	Result
Screens render	Chromium, 7 screens × 1366 and 1920 px	No console errors
Live answers	All 10 demo clips posted to the live server; 4 also played through the Capture screen	9/9 reliable clips correct (66–97 %); look-alike clip routed to Not sure
Same answers as the evaluation	Server classification vs offline result, same clips and model	8 of 8 clips agree
Vocabulary restriction	50-word answers vs all-262-word answers	Matches the evaluation's 85.9 % setting; 262 words would give 72.0 %
GAN front end in the interface	180p copy of a demo clip, SR-GAN enabled	Read correctly
Latency	Desktop CPU, six views, end to end	5–9 s per clip

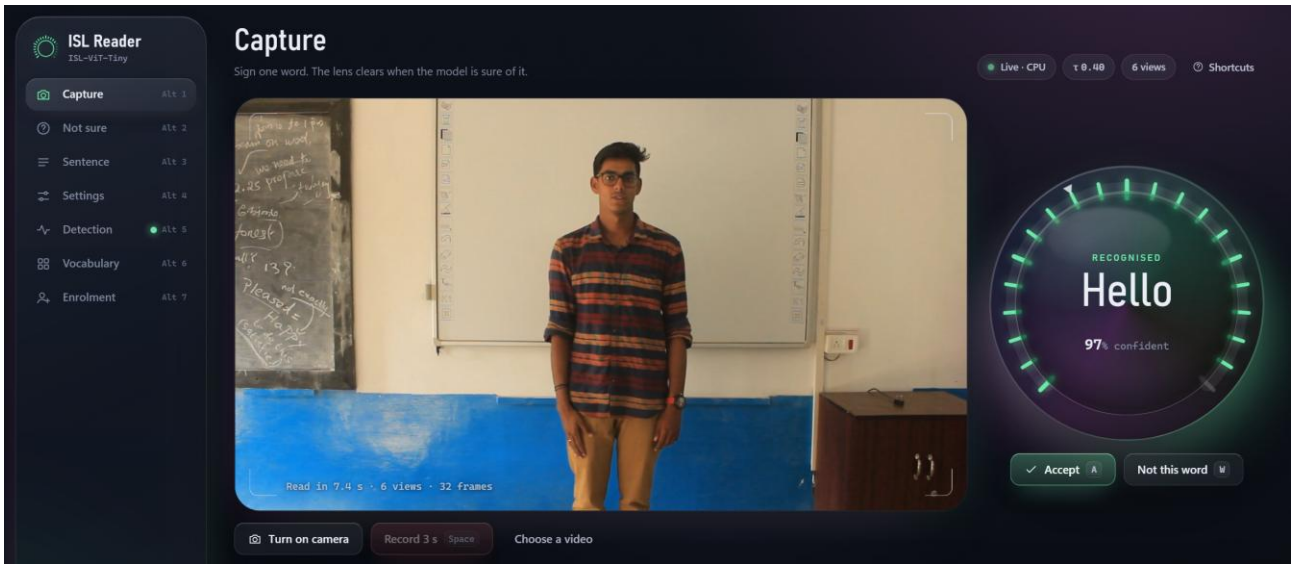


Fig. 14 Capture screen with the live model reading a held-out demonstration clip.

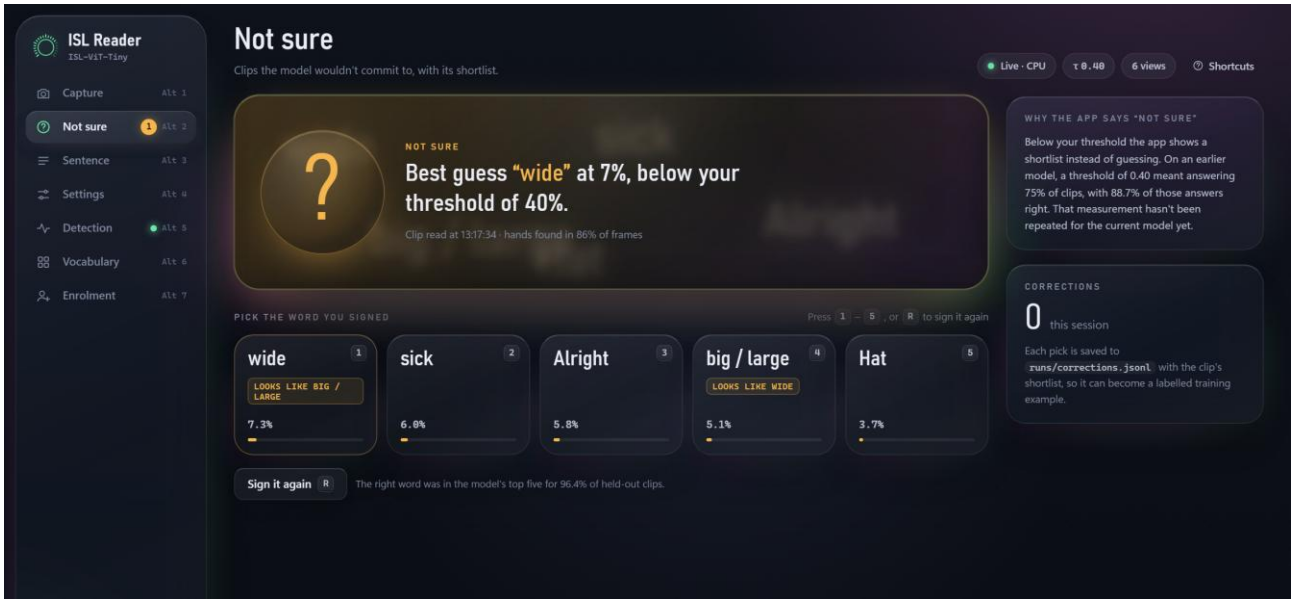


Fig. 15 Not-sure screen for the look-alike demonstration clip: the model's top probability is below the threshold, so the user picks from the shortlist.

4.7 Limitations

INCLUDE was recorded in one studio in one city, sessions are inferred from camera counters rather than signer labels, and the classifier has not been shown to transfer to another corpus. The super-resolution study degrades studio video synthetically; real phone or glasses video, with motion blur and inter-frame compression, may behave differently. The confidence curve shown in the interface comes from an earlier model. Latency was measured on a desktop CPU, not on wearable hardware, and the system recognises isolated words, not continuous signing. Finally, version 3 of the generator was designed after the version 1 and 2 results were known, so it is not part of the pre-registered comparison; we report it in full for that reason.

5 Conclusion

Measured on recording sessions it never saw, a 3.89 M-parameter factorised transformer recognises 50 ISL words at 85.9 % top-1 from a 1.97 MiB file, and the same model scores over 20 points higher when the split leaks sessions. Learned super-resolution in front of the hand detector recovers most of the accuracy lost on low-resolution video, up to 14.5 points at 180p, but the adversarial part of the GAN adds little beyond a non-adversarial upscaler, and a generator trained to preserve hand landmarks improved detection statistics without improving classification. After super-resolution the remaining loss lies mainly in landmark placement. The interface ties these models together and reproduces the offline evaluation's answers. The next steps are to fine-tune the classifier on the task-aware front end, re-measure the confidence curve for the shipped model, fix hand detection for tightly framed video such as CISLR, and test on recordings from new signers through the Enrolment screen and on wearable hardware.

References

1. Lamaakal I, Yahyati C, Maleh Y, El Makkaoui K, et al. (2026) An explainable hybrid CNN–transformer model for sign language recognition on edge devices using adaptive fusion and knowledge distillation. Sci Rep 16:7143. https://doi.org/10.1038/s41598-026-38478-8
2. Venkatappa U, N. K. S., N. K. S., Bhat PP, Srinivas P (2026) Dynamic Kannada sign language recognition on resource constrained devices. Sci Rep 16:11186. https://doi.org/10.1038/s41598-026-40181-7
3. Ashfaq U, Wang Q, Merabet B, Zhang J (2026) SignViT: an enhanced vision transformer framework for attention-based sign language hand gesture recognition. Biomed Signal Process Control 112(Part B):108602. https://doi.org/10.1016/j.bspc.2025.108602
4. Rubaiyeat HA, Youssouf N, Hasan MK, et al. (2026) Transformer-based word level Bangla sign language recognition using relative quantization encoding. Sci Rep 16:25259. https://doi.org/10.1038/s41598-026-53835-3
5. Abdi A, Barzegar Touchahi F (2026) Underwater sign language recognition using convolutional and transformer models: a comparative study in clear and murky conditions. Signal Image Video Process 20:297. https://doi.org/10.1007/s11760-026-05314-5
6. Nedungadi P, et al. (2025) Vision transformer-powered conversational agent for real-time Indian Sign Language e-governance accessibility. Sci Rep 15:33055. https://doi.org/10.1038/s41598-025-12667-3

7. Govindharajalu Kaliyaperumal V, et al. (2025) A deep neural network framework for dynamic two-handed Indian Sign Language recognition in hearing and speech-impaired communities. Sensors 25(12):3652. https://doi.org/10.3390/s25123652

8. Shawon JAB, et al. (2026) A comparative analysis of video vision transformers on word-level sign language datasets. PLOS One 21(2):e0341909. https://doi.org/10.1371/journal.pone.0341909

9. Damdoo R, et al. (2026) End-to-end sentence-level Indian sign language translation with ISH-NEWS dataset and transformer model. Sci Rep. https://doi.org/10.1038/s41598-026-60893-0

10. Damdoo R, et al. (2025) An integrative survey on Indian sign language recognition and translation. IET Image Process 19(1):e70000. https://doi.org/10.1049/ipr2.70000

11. Alyami S, et al. (2026) Isharah: a large-scale multi-scene dataset for continuous sign language recognition. IEEE Trans Multimedia 28:5050–5058. https://doi.org/10.1109/TMM.2026.3664959

12. Chalotonpised J, et al. (2026) TSL-ONE-S: a real-world Thai Sign Language dataset with deep learning benchmarks. IEEE Access 14:37845–37870. https://doi.org/10.1109/ACCESS.2026.3670970

13. Arib SH, et al. (2025) SignFormer-GCN: continuous sign language translation using spatio-temporal graph convolutional networks. PLOS One 20(2):e0316298. https://doi.org/10.1371/journal.pone.0316298

14. Said Y, et al. (2025) Adaptive transformer-based deep learning framework for continuous sign language recognition and translation. Mathematics 13(6):909. https://doi.org/10.3390/math13060909

15. López-Nava IH, et al. (2026) A comprehensive dataset of static and dynamic signs for the Mexican Sign Language alphabet. Data Brief 66:112887. https://doi.org/10.1016/j.dib.2026.112887

16. Sridhar A, Ganesan RG, Kumar P, Khapra M (2020) INCLUDE: a large scale dataset for Indian Sign Language recognition. In: Proceedings of the 28th ACM International Conference on Multimedia, pp 1366–1375. https://doi.org/10.1145/3394171.3413528

17. Joshi A, et al. (2022) CISLR: Corpus for Indian Sign Language recognition. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing

18. Joshi A, Mohanty R, Kanakanti M, Mangla A, Choudhary S, Barbate M, Modi A (2024) iSign: a benchmark for Indian Sign Language processing. In: Findings of the Association for Computational Linguistics: ACL 2024

19. Dosovitskiy A, et al. (2021) An image is worth 16x16 words: transformers for image recognition at scale. In: International Conference on Learning Representations

20. Touvron H, Cord M, Douze M, Massa F, Sablayrolles A, Jégou H (2021) Training data-efficient image transformers and distillation through attention. In: Proceedings of the 38th International Conference on Machine Learning, PMLR 139, pp 10347–10357

21. Arnab A, Dehghani M, Heigold G, Sun C, Lučić M, Schmid C (2021) ViViT: a video vision transformer. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp 6836–6846

22. Xie Z, Zhang Z, Cao Y, Lin Y, Bao J, Yao Z, Dai Q, Hu H (2022) SimMIM: a simple framework for masked image modeling. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 9653–9663

23. Lugaresi C, et al. (2019) MediaPipe: a framework for building perception pipelines. arXiv:1906.08172

24. Jacob B, Kligys S, Chen B, Zhu M, Tang M, Howard A, Adam H, Kalenichenko D (2018) Quantization and training of neural networks for efficient integer-arithmetic-only inference. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 2704–2713

25. Bengio Y, Léonard N, Courville A (2013) Estimating or propagating gradients through stochastic neurons for conditional computation. arXiv:1308.3432

26. McNemar Q (1947) Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika 12(2):153–157

27. Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. In: Advances in Neural Information Processing Systems 27, pp 2672–2680

28. Ledig C, Theis L, Huszár F, et al. (2017) Photo-realistic single image super-resolution using a generative adversarial network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 4681–4690

29. Wang X, Yu K, Wu S, Gu J, Liu Y, Dong C, Qiao Y, Loy CC (2018) ESRGAN: enhanced super-resolution generative adversarial networks. In: Computer Vision – ECCV 2018 Workshops

30. Wang X, Xie L, Dong C, Shan Y (2021) Real-ESRGAN: training real-world blind super-resolution with pure synthetic data. In: Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops, pp 1905–1914

31. Schönfeld E, Schiele B, Khoreva A (2020) A U-Net based discriminator for generative adversarial networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 8207–8216

32. Jolicoeur-Martineau A (2019) The relativistic discriminator: a key element missing from standard GAN. In: International Conference on Learning Representations

33. Johnson J, Alahi A, Fei-Fei L (2016) Perceptual losses for real-time style transfer and super-resolution. In: Computer Vision – ECCV 2016, pp 694–711

34. Simonyan K, Zisserman A (2015) Very deep convolutional networks for large-scale image recognition. In: International Conference on Learning Representations

35. Miyato T, Kataoka T, Koyama M, Yoshida Y (2018) Spectral normalization for generative adversarial networks. In: International Conference on Learning Representations

36. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 770–778
37. Sun X, Xiao B, Wei F, Liang S, Wei Y (2018) Integral human pose regression. In: Computer Vision – ECCV 2018, pp 529–545